SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	Manoj M	**Reg. No.:**	192511060
Section / Batch:	B.E CSE	**Date:**	23.07.2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Q51. Application: Smart Agriculture Monitoring System**

A smart agriculture system processes field sensor data using the recurrence $T(n) = 5T(n/2) + n^2$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how performance changes as the number of sensors increases. Interpret efficiency using asymptotic notation.

Q52. Application: Virtual Reality Rendering System

A virtual reality system renders scenes using the recurrence $T(n) = 6T(n/3) + n \log n$. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the selected case. Analyze how rendering performance scales with increasing scene complexity. Interpret efficiency using asymptotic notation.

Code of Conduct

I _Manoj M(192511060)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

Smart Agriculture Monitoring System

Scenario-Based Execution

Qn) 51

Problem

A smart agriculture system processes field sensor data using the recurrence:

T(n)=5T(n/2)+n2T(n)=5T(n/2)+n^2T(n)=5T(n/2)+n2

Step 1: Identify the Parameters

From the recurrence,

- **a = 5** (number of recursive calls)
- **b = 2** (input size is divided by 2)
- **f(n) = n²** (work done outside the recursive calls)

Step 2: Compute nlogban^{\log_b a}nlogb a

nlog25n^{\log_2 5}nlog2 5

Using logarithm,

log25≈2.32\log_2 5 \approx 2.32log2 5≈2.32

Therefore,

nlog25=n2.32n^{\log_2 5}=n^{2.32}nlog2 5=n2.32

Step 3: Compare f(n)f(n)f(n) and nlogban^{\log_b a}nlogb a

We compare

- f(n)=n2f(n)=n^2f(n)=n2
- nlog25=n2.32n^{\log_2 5}=n^{2.32}nlog2 5=n2.32

Since

n2<n2.32n^2<n^{2.32}n2<n2.32

or equivalently,

f(n)=O(n2.32−ε)f(n)=O(n^{2.32-\varepsilon})f(n)=O(n2.32−ε)

for some ε>0\varepsilon>0ε>0, the recursive part grows faster.

Hence, **Master Theorem Case 1** applies.

Step 4: Determine the Time Complexity

According to Master Theorem Case 1,

T(n)=Θ(nlog25)T(n)=\Theta(n^{\log_2 5})T(n)=Θ(nlog2 5)

Therefore,

T(n)=Θ(n2.32)\boxed{T(n)=\Theta(n^{2.32})}T(n)=Θ(n2.32)

Execution Analysis

- Initially, all sensor data is divided into **2 smaller groups**.
- Each group generates **5 recursive processing tasks**.
- Every level also performs n2n^2n2 operations for processing the current sensor data.
- As the number of sensors increases, recursive calls dominate the total work.

Conclusion: The algorithm scales approximately as n2.32n^{2.32}n2.32, which is faster than cubic time but slower than quadratic time.

Virtual Reality Rendering System

Scenario-Based Execution

Qn) 52

Problem

A virtual reality rendering system follows

T(n)=6T(n/3)+nlognT(n)=6T(n/3)+n\log nT(n)=6T(n/3)+nlogn

Step 1: Identify the Parameters

From the recurrence,

- **a = 6**
- **b = 3**
- **f(n)=n\log n**

Step 2: Compute nlogban^{\log_b a}nlogb a

nlog36n^{\log_3 6}nlog3 6

Calculate,

log36≈1.63\log_3 6\approx1.63log3 6≈1.63

Therefore,

nlog36=n1.63n^{\log_3 6}=n^{1.63}nlog3 6=n1.63

Step 3: Compare f(n)f(n)f(n) and nlogban^{\log_b a}nlogb a

Compare

- f(n)=nlognf(n)=n\log nf(n)=nlogn
- n1.63n^{1.63}n1.63

Since

nlogn<n1.63n\log n<n^{1.63}nlogn<n1.63

or

f(n)=O(n1.63−ε)f(n)=O(n^{1.63-\varepsilon})f(n)=O(n1.63−ε)

for some ε>0\varepsilon>0ε>0, the recursive work dominates.

Therefore, **Master Theorem Case 1** applies.

Step 4: Determine the Time Complexity

Using Master Theorem Case 1,

T(n)=Θ(nlog36)T(n)=\Theta(n^{\log_3 6})T(n)=Θ(nlog3 6)

Hence,

T(n)=Θ(n1.63)\boxed{T(n)=\Theta(n^{1.63})}T(n)=Θ(n1.63)

Execution Analysis

- The rendering engine divides the scene into **3 smaller parts**.
- Each part generates **6 recursive rendering tasks**.
- At every level, an additional nlogn\log nnlogn operations are required for rendering calculations.
- As scene complexity increases, recursive rendering becomes the dominant cost.

Conclusion: The rendering time grows approximately as n1.63n^{1.63}n1.63. This is more efficient than quadratic time, allowing the system to handle larger scenes relatively well.